

```
In [2]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
data = pd.read_csv("Amazon1.csv")
print(data)
```

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	\
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	
...	...	...	...	...	...	
99995	ORD0099996	2023-03-07	CUST001356	Karan Joshi	P00047	
99996	ORD0099997	2021-11-24	CUST031254	Sunita Kapoor	P00046	
99997	ORD0099998	2023-04-29	CUST012579	Aman Gupta	P00030	
99998	ORD0099999	2021-11-01	CUST026243	Simran Gupta	P00046	
99999	ORD0100000	2021-12-04	CUST029492	Sunita Reddy	P00019	

	ProductName	Category	Brand	Quantity	\
0	Drone Mini	Books	BrightLux	3	
1	Microphone	Home & Kitchen	UrbanStyle	1	
2	Power Bank 20000mAh	Clothing	UrbanStyle	3	
3	Webcam Full HD	Home & Kitchen	Zenith	5	
4	T-Shirt	Clothing	KiddoFun	2	
...	...	...	...	...	
99995	Memory Card 128GB	Electronics	Apex	2	
99996	Car Charger	Sports & Outdoors	Apex	5	
99997	Dress Shirt	Sports & Outdoors	BrightLux	4	
99998	Car Charger	Sports & Outdoors	HomeEase	1	
99999	LED Desk Lamp	Home & Kitchen	CoreTech	5	

	UnitPrice	Discount	Tax	ShippingCost	TotalAmount	\
0	106.59	0.00	0.00	0.09	319.86	
1	251.37	0.05	19.10	1.74	259.64	
2	35.03	0.10	7.57	5.91	108.06	
3	33.58	0.15	11.42	5.53	159.66	
4	515.64	0.25	38.67	9.23	821.36	
...	...	...	...	...	...	
99995	492.34	0.00	78.77	2.75	1066.20	
99996	449.30	0.00	179.72	6.07	2432.29	
99997	232.40	0.00	74.37	12.43	1016.40	
99998	294.05	0.00	23.52	13.09	330.66	
99999	166.70	0.05	63.35	3.34	858.52	

	PaymentMethod	OrderStatus	City	State	Country	\
0	Debit Card	Delivered	Washington	DC	India	
1	Amazon Pay	Delivered	Fort Worth	TX	United States	
2	Debit Card	Delivered	Austin	TX	United States	
3	Cash on Delivery	Delivered	Charlotte	NC	India	
4	Credit Card	Cancelled	San Antonio	TX	Canada	
...	...	...	...	...	...	
99995	UPI	Delivered	Jacksonville	FL	India	
99996	Credit Card	Delivered	San Jose	CA	United States	
99997	Cash on Delivery	Delivered	Indianapolis	IN	United States	
99998	Debit Card	Delivered	Charlotte	NC	United States	
99999	Debit Card	Delivered	New York	NY	United States	

	SellerID
0	SELL01967
1	SELL01298
2	SELL00908

```
3      SELL01164
4      SELL01411
...      ...
99995  SELL00041
99996  SELL01449
99997  SELL00028
99998  SELL00324
99999  SELL00761
```

```
[100000 rows x 20 columns]
```

```
In [3]: df = pd.DataFrame(data)
        print(df)
```

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	\
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	
...	...	...	...	...	...	
99995	ORD0099996	2023-03-07	CUST001356	Karan Joshi	P00047	
99996	ORD0099997	2021-11-24	CUST031254	Sunita Kapoor	P00046	
99997	ORD0099998	2023-04-29	CUST012579	Aman Gupta	P00030	
99998	ORD0099999	2021-11-01	CUST026243	Simran Gupta	P00046	
99999	ORD0100000	2021-12-04	CUST029492	Sunita Reddy	P00019	

	ProductName	Category	Brand	Quantity	\
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2	Power Bank 20000mAh	Clothing	UrbanStyle	3	
3	Webcam Full HD	Home & Kitchen	Zenith	5	
4	T-Shirt	Clothing	KiddoFun	2	
...	...	...	...	...	
99995	Memory Card 128GB	Electronics	Apex	2	
99996	Car Charger	Sports & Outdoors	Apex	5	
99997	Dress Shirt	Sports & Outdoors	BrightLux	4	
99998	Car Charger	Sports & Outdoors	HomeEase	1	
99999	LED Desk Lamp	Home & Kitchen	CoreTech	5	

	UnitPrice	Discount	Tax	ShippingCost	TotalAmount	\
0	106.59	0.00	0.00	0.09	319.86	
1	251.37	0.05	19.10	1.74	259.64	
2	35.03	0.10	7.57	5.91	108.06	
3	33.58	0.15	11.42	5.53	159.66	
4	515.64	0.25	38.67	9.23	821.36	
...	...	...	...	...	...	
99995	492.34	0.00	78.77	2.75	1066.20	
99996	449.30	0.00	179.72	6.07	2432.29	
99997	232.40	0.00	74.37	12.43	1016.40	
99998	294.05	0.00	23.52	13.09	330.66	
99999	166.70	0.05	63.35	3.34	858.52	

	PaymentMethod	OrderStatus	City	State	Country	\
0	Debit Card	Delivered	Washington	DC	India	
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2	Debit Card	Delivered	Austin	TX	United States	
3	Cash on Delivery	Delivered	Charlotte	NC	India	
4	Credit Card	Cancelled	San Antonio	TX	Canada	
...	...	...	...	...	...	
99995	UPI	Delivered	Jacksonville	FL	India	
99996	Credit Card	Delivered	San Jose	CA	United States	
99997	Cash on Delivery	Delivered	Indianapolis	IN	United States	
99998	Debit Card	Delivered	Charlotte	NC	United States	
99999	Debit Card	Delivered	New York	NY	United States	

	SellerID
0	SELL01967
1	SELL01298
2	SELL00908

3 SELL01164
4 SELL01411
... ...
99995 SELL00041
99996 SELL01449
99997 SELL00028
99998 SELL00324
99999 SELL00761

[100000 rows x 20 columns]

```
In [4]: df.head(10)
```

Out[4]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Category
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	Drone Mini	Box
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	Microphone	Home Kitch
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	Power Bank 20000mAh	Clothi
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	Webcam Full HD	Home Kitch
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	T-Shirt	Clothi
5	ORD0000006	2022-12-31	CUST042705	Karan Sharma	P00023	Cookware Set	Box
6	ORD0000007	2024-09-20	CUST037667	Aarav Verma	P00030	Dress Shirt	Clothi
7	ORD0000008	2022-11-10	CUST031165	Rohit Kumar	P00028	Jeans	Toys Gam
8	ORD0000009	2024-06-26	CUST026965	Aman Kapoor	P00031	Kids Toy Car	Sports Outdo
9	ORD0000010	2020-05-01	CUST029472	Aarav Reddy	P00001	Wireless Earbuds	Clothi

```
In [5]: df.tail(10)
```

Out[5]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Category
99990	ORD0099991	2024-05-29	CUST019886	Sneha Kapoor	P00001	Wireless Earbuds	Electronics
99991	ORD0099992	2021-05-08	CUST005627	Kabir Reddy	P00009	Portable SSD 1TB	Storage
99992	ORD0099993	2021-07-17	CUST030861	Simran Verma	P00017	Electric Kettle	Home Appliances
99993	ORD0099994	2023-05-04	CUST045239	Mohit Sharma	P00029	T-Shirt	Clothing
99994	ORD0099995	2022-01-09	CUST028035	Kabir Kapoor	P00011	Fitness Band	Wearables
99995	ORD0099996	2023-03-07	CUST001356	Karan Joshi	P00047	Memory Card 128GB	Storage
99996	ORD0099997	2021-11-24	CUST031254	Sunita Kapoor	P00046	Car Charger	Automotive
99997	ORD0099998	2023-04-29	CUST012579	Aman Gupta	P00030	Dress Shirt	Clothing
99998	ORD0099999	2021-11-01	CUST026243	Simran Gupta	P00046	Car Charger	Automotive
99999	ORD0100000	2021-12-04	CUST029492	Sunita Reddy	P00019	LED Desk Lamp	Home Appliances

In [6]: df.shape

Out[6]: (100000, 20)

In [7]: df.columns

Out[7]: Index(['OrderID', 'OrderDate', 'CustomerID', 'CustomerName', 'ProductID', 'ProductName', 'Category', 'Brand', 'Quantity', 'UnitPrice', 'Discount', 'Tax', 'ShippingCost', 'TotalAmount', 'PaymentMethod', 'OrderStatus', 'City', 'State', 'Country', 'SellerID'], dtype='str')

In [8]: df.dtypes

```
Out[8]: OrderID          str
        OrderDate       str
        CustomerID      str
        CustomerName    str
        ProductID       str
        ProductName     str
        Category        str
        Brand           str
        Quantity        int64
        UnitPrice       float64
        Discount        float64
        Tax             float64
        ShippingCost    float64
        TotalAmount     float64
        PaymentMethod   str
        OrderStatus     str
        City            str
        State           str
        Country         str
        SellerID        str
        dtype: object
```

```
In [9]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 20 columns):
#   Column          Non-Null Count  Dtype
---  -
0   OrderID         100000 non-null  str
1   OrderDate       100000 non-null  str
2   CustomerID      100000 non-null  str
3   CustomerName    100000 non-null  str
4   ProductID       100000 non-null  str
5   ProductName     100000 non-null  str
6   Category        100000 non-null  str
7   Brand           100000 non-null  str
8   Quantity        100000 non-null  int64
9   UnitPrice       100000 non-null  float64
10  Discount        100000 non-null  float64
11  Tax             100000 non-null  float64
12  ShippingCost    100000 non-null  float64
13  TotalAmount     100000 non-null  float64
14  PaymentMethod   100000 non-null  str
15  OrderStatus     100000 non-null  str
16  City            100000 non-null  str
17  State           100000 non-null  str
18  Country         100000 non-null  str
19  SellerID        100000 non-null  str
dtypes: float64(5), int64(1), str(14)
memory usage: 15.3 MB
```

```
In [10]: df.describe()
```

Out[10]:

	Quantity	UnitPrice	Discount	Tax	ShippingCost	TotalA
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.
mean	3.001400	302.905748	0.074226	68.468902	7.406660	918.
std	1.413548	171.840797	0.082583	74.131180	4.324057	724.
min	1.000000	5.000000	0.000000	0.000000	0.000000	4.
25%	2.000000	154.190000	0.000000	15.920000	3.680000	340.
50%	3.000000	303.070000	0.050000	45.250000	7.300000	714.
75%	4.000000	451.500000	0.100000	96.060000	11.150000	1349.
max	5.000000	599.990000	0.300000	538.460000	15.000000	3534.



In [11]: `df.isnull().sum()`

Out[11]:

OrderID	0
OrderDate	0
CustomerID	0
CustomerName	0
ProductID	0
ProductName	0
Category	0
Brand	0
Quantity	0
UnitPrice	0
Discount	0
Tax	0
ShippingCost	0
TotalAmount	0
PaymentMethod	0
OrderStatus	0
City	0
State	0
Country	0
SellerID	0
dtype:	int64

In [12]: `df.duplicated()`

```
Out[12]: 0      False
         1      False
         2      False
         3      False
         4      False
         ...
        99995    False
        99996    False
        99997    False
        99998    False
        99999    False
        Length: 100000, dtype: bool
```

```
In [13]: df.duplicated().sum()
```

```
Out[13]: np.int64(0)
```

```
In [18]: df["OrderDate"] = pd.to_datetime(df["OrderDate"])
```

```
In [19]: df.dtypes
```

```
Out[19]: OrderID          str
         OrderDate      datetime64[us]
         CustomerID     str
         CustomerName    str
         ProductID       str
         ProductName     str
         Category        str
         Brand           str
         Quantity        int64
         UnitPrice       float64
         Discount        float64
         Tax             float64
         ShippingCost    float64
         TotalAmount     float64
         PaymentMethod    str
         OrderStatus     str
         City            str
         State           str
         Country         str
         SellerID        str
         dtype: object
```

```
In [20]: df["OrderMonth"] = df["OrderDate"].dt.month
```

```
In [21]: df.dtypes
```

```
Out[21]: OrderID          str
OrderDate      datetime64[us]
CustomerID     str
CustomerName   str
ProductID      str
ProductName     str
Category       str
Brand          str
Quantity       int64
UnitPrice      float64
Discount       float64
Tax            float64
ShippingCost   float64
TotalAmount    float64
PaymentMethod  str
OrderStatus    str
City           str
State          str
Country        str
SellerID       str
OrderMonth     int32
dtype: object
```

```
In [22]: df[["OrderDate", "OrderMonth"]].head()
```

```
Out[22]:
```

	OrderDate	OrderMonth
0	2023-01-31	1
1	2023-12-30	12
2	2022-05-10	5
3	2023-07-18	7
4	2023-02-04	2

```
In [23]: df["OrderYear"] = df["OrderDate"].dt.year
```

```
In [24]: df[["OrderDate", "OrderYear"]].head()
```

```
Out[24]:
```

	OrderDate	OrderYear
0	2023-01-31	2023
1	2023-12-30	2023
2	2022-05-10	2022
3	2023-07-18	2023
4	2023-02-04	2023

```
In [25]: df["MonthName"] = df["OrderDate"].dt.month_name()
```

```
In [26]: df[["OrderDate", "MonthName"]].head()
```

```
Out[26]:
```

	OrderDate	MonthName
0	2023-01-31	January
1	2023-12-30	December
2	2022-05-10	May
3	2023-07-18	July
4	2023-02-04	February

```
In [27]: df["DayName"] = df["OrderDate"].dt.day_name()
```

```
In [28]: df[["OrderDate", "DayName"]].head()
```

```
Out[28]:
```

	OrderDate	DayName
0	2023-01-31	Tuesday
1	2023-12-30	Saturday
2	2022-05-10	Tuesday
3	2023-07-18	Tuesday
4	2023-02-04	Saturday

```
In [29]: df[["OrderDate", "DayName", "MonthName", "OrderYear"]].head(10)
```

```
Out[29]:
```

	OrderDate	DayName	MonthName	OrderYear
0	2023-01-31	Tuesday	January	2023
1	2023-12-30	Saturday	December	2023
2	2022-05-10	Tuesday	May	2022
3	2023-07-18	Tuesday	July	2023
4	2023-02-04	Saturday	February	2023
5	2022-12-31	Saturday	December	2022
6	2024-09-20	Friday	September	2024
7	2022-11-10	Thursday	November	2022
8	2024-06-26	Wednesday	June	2024
9	2020-05-01	Friday	May	2020

```
In [30]: df["Quarter"] = df["OrderDate"].dt.quarter
```

```
In [31]: df[["OrderDate", "Quarter"]].head()
```

```
Out[31]:
```

	OrderDate	Quarter
0	2023-01-31	1
1	2023-12-30	4
2	2022-05-10	2
3	2023-07-18	3
4	2023-02-04	1

```
In [32]: df[["OrderDate", "DayName", "MonthName", "OrderYear", "Quarter"]].head(10)
```

```
Out[32]:
```

	OrderDate	DayName	MonthName	OrderYear	Quarter
0	2023-01-31	Tuesday	January	2023	1
1	2023-12-30	Saturday	December	2023	4
2	2022-05-10	Tuesday	May	2022	2
3	2023-07-18	Tuesday	July	2023	3
4	2023-02-04	Saturday	February	2023	1
5	2022-12-31	Saturday	December	2022	4
6	2024-09-20	Friday	September	2024	3
7	2022-11-10	Thursday	November	2022	4
8	2024-06-26	Wednesday	June	2024	2
9	2020-05-01	Friday	May	2020	2

```
In [33]: df["PriceCategory"] = pd.cut(df["UnitPrice"], bins = [0,200,400,600], labels = ["Lo
```

```
In [34]: df[["UnitPrice", "PriceCategory"]].head()
```

```
Out[34]:
```

	UnitPrice	PriceCategory
0	106.59	Low
1	251.37	Medium
2	35.03	Low
3	33.58	Low
4	515.64	High

```
In [35]: df["DiscountPercent"] = df["Discount"] * 100
```

In [36]: `df[["Discount", "DiscountPercent"]].head()`

Out[36]:

	Discount	DiscountPercent
0	0.00	0.0
1	0.05	5.0
2	0.10	10.0
3	0.15	15.0
4	0.25	25.0

In [37]: `df[["OrderDate", "OrderMonth", "OrderYear", "MonthName", "DayName", "Quarter", "PriceCategory"]].head()`

Out[37]:

	OrderDate	OrderMonth	OrderYear	MonthName	DayName	Quarter	PriceCategory
0	2023-01-31	1	2023	January	Tuesday	1	Low
1	2023-12-30	12	2023	December	Saturday	4	Medium
2	2022-05-10	5	2022	May	Tuesday	2	Low
3	2023-07-18	7	2023	July	Tuesday	3	Low
4	2023-02-04	2	2023	February	Saturday	1	High
5	2022-12-31	12	2022	December	Saturday	4	High
6	2024-09-20	9	2024	September	Friday	3	Medium
7	2022-11-10	11	2022	November	Thursday	4	Medium
8	2024-06-26	6	2024	June	Wednesday	2	Low
9	2020-05-01	5	2020	May	Friday	2	Medium

In [38]: `df["Category"] = df["Category"].str.title()`

In [39]: `df["Category"].head(10)`

```
Out[39]: 0      Books
1  Home & Kitchen
2      Clothing
3  Home & Kitchen
4      Clothing
5      Books
6      Clothing
7  Toys & Games
8  Sports & Outdoors
9      Clothing
Name: Category, dtype: str
```

```
In [40]: df["Brand"] = df["Brand"].str.upper()
df["Brand"].head(10)
```

```
Out[40]: 0    BRIGHTLUX
1    URBANSTYLE
2    URBANSTYLE
3      ZENITH
4    KIDDOFUN
5    READMORE
6    URBANSTYLE
7    KIDDOFUN
8      APEX
9      APEX
Name: Brand, dtype: str
```

```
In [41]: df = df.sort_values(by = "TotalAmount", ascending=False)
df[["ProductName", "TotalAmount"]].head(10)
```

```
Out[41]:
```

	ProductName	TotalAmount
48442	Puzzle 1000pc	3534.98
81886	Wireless Charger	3521.85
4196	T-Shirt	3514.24
1929	Running Shoes	3513.65
58204	Kids Toy Car	3513.05
46764	Desk Organizer	3511.95
40146	Bluetooth Speaker	3510.23
20578	Instant Pot	3509.30
95429	Gaming Mouse	3499.15
51547	Action Camera	3498.13

```
In [42]: high_sales = df[df["TotalAmount"] > 100]
high_sales.head(10)
```

Out[42]:

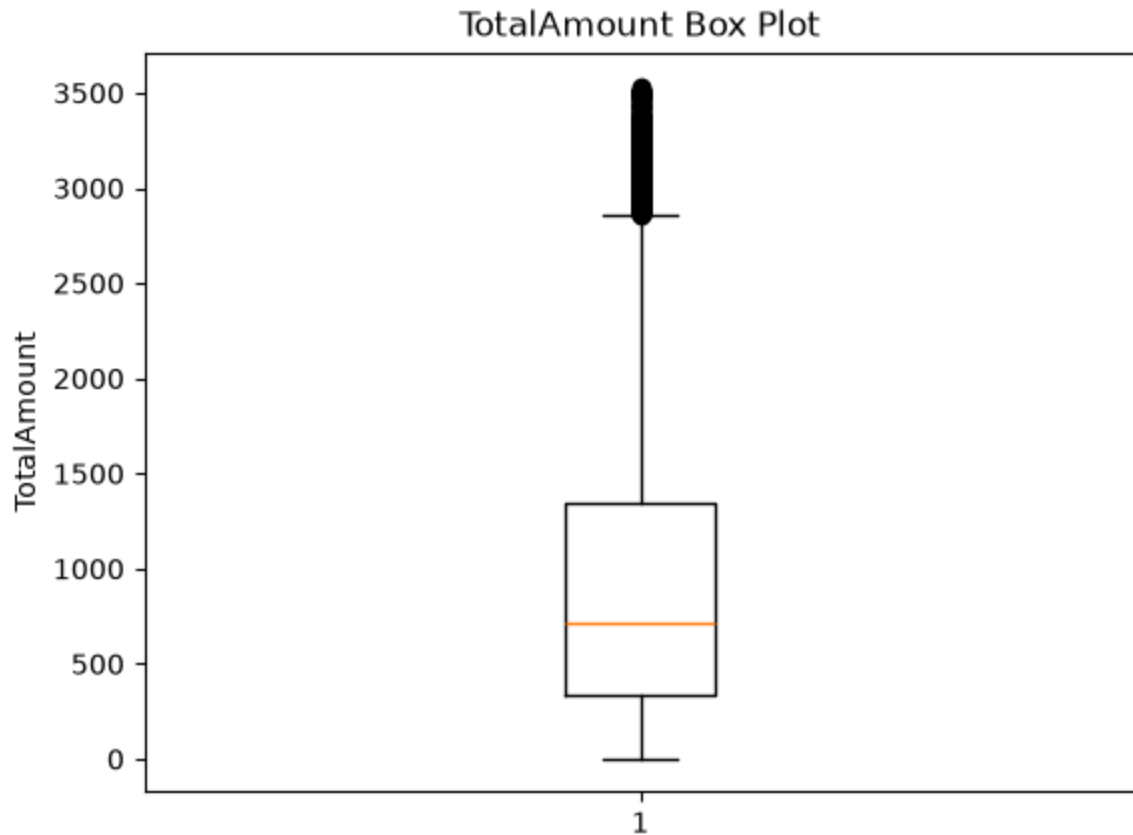
	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	C
48442	ORD0048443	2020-01-12	CUST010794	Anjali Patel	P00033	Puzzle 1000pc	I
81886	ORD0081887	2020-09-26	CUST031626	Vivaan Joshi	P00048	Wireless Charger	
4196	ORD0004197	2021-02-22	CUST022868	Kabir Gupta	P00029	T-Shirt	I
1929	ORD0001930	2023-08-18	CUST031209	Pooja Patel	P00025	Running Shoes	
58204	ORD0058205	2021-10-19	CUST033287	Priya Gupta	P00031	Kids Toy Car	I
46764	ORD0046765	2024-04-13	CUST044400	Priya Kapoor	P00034	Desk Organizer	I
40146	ORD0040147	2024-10-07	CUST001991	Pooja Mehta	P00002	Bluetooth Speaker	
20578	ORD0020579	2023-09-24	CUST016754	Sneha Sharma	P00015	Instant Pot	S C
95429	ORD0095430	2023-03-08	CUST001586	Sunita Joshi	P00006	Gaming Mouse	
51547	ORD0051548	2024-03-22	CUST024376	Vihaan Gupta	P00013	Action Camera	I

10 rows × 27 columns



In [43]:

```
plt.boxplot(df["TotalAmount"])
plt.title("TotalAmount Box Plot")
plt.ylabel("TotalAmount")
plt.show()
```



```
In [44]: q1 = df["TotalAmount"].quantile(0.25)
q3 = df["TotalAmount"].quantile(0.75)
IQR = q3 - q1
print(q1, q3, IQR)

340.89 1349.7649999999999 1008.8749999999999
```

```
In [45]: lower_limit = q1 - 1.5 * IQR
upper_limit = q3 + 1.5 * IQR
print(lower_limit)
print(upper_limit)

-1172.4224999999997
2863.0774999999994
```

```
In [46]: outliers = df[(df["TotalAmount"] < lower_limit) | (df["TotalAmount"] > upper_limit)]
outliers.head(10)
```

Out[46]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	C
48442	ORD0048443	2020-01-12	CUST010794	Anjali Patel	P00033	Puzzle 1000pc	I
81886	ORD0081887	2020-09-26	CUST031626	Vivaan Joshi	P00048	Wireless Charger	
4196	ORD0004197	2021-02-22	CUST022868	Kabir Gupta	P00029	T-Shirt	I
1929	ORD0001930	2023-08-18	CUST031209	Pooja Patel	P00025	Running Shoes	
58204	ORD0058205	2021-10-19	CUST033287	Priya Gupta	P00031	Kids Toy Car	I
46764	ORD0046765	2024-04-13	CUST044400	Priya Kapoor	P00034	Desk Organizer	I
40146	ORD0040147	2024-10-07	CUST001991	Pooja Mehta	P00002	Bluetooth Speaker	
20578	ORD0020579	2023-09-24	CUST016754	Sneha Sharma	P00015	Instant Pot	S C
95429	ORD0095430	2023-03-08	CUST001586	Sunita Joshi	P00006	Gaming Mouse	
51547	ORD0051548	2024-03-22	CUST024376	Vihaan Gupta	P00013	Action Camera	I

10 rows × 27 columns

In [47]: len(outliers)

Out[47]: 1360

In [48]: df.shape

Out[48]: (100000, 27)

In [49]: df.isnull().sum()

```
Out[49]: OrderID      0
         OrderDate    0
         CustomerID   0
         CustomerName 0
         ProductID    0
         ProductName  0
         Category     0
         Brand         0
         Quantity     0
         UnitPrice    0
         Discount     0
         Tax          0
         ShippingCost 0
         TotalAmount  0
         PaymentMethod 0
         OrderStatus  0
         City         0
         State        0
         Country      0
         SellerID     0
         OrderMonth   0
         OrderYear    0
         MonthName    0
         DayName      0
         Quarter      0
         PriceCategory 0
         DiscountPercent 0
         dtype: int64
```

```
In [50]: df.duplicated().sum()
```

```
Out[50]: np.int64(0)
```

```
In [51]: df.dtypes
```

```
Out[51]: OrderID          str
OrderDate      datetime64[us]
CustomerID     str
CustomerName   str
ProductID      str
ProductName    str
Category       str
Brand          str
Quantity       int64
UnitPrice      float64
Discount       float64
Tax            float64
ShippingCost   float64
TotalAmount    float64
PaymentMethod  str
OrderStatus    str
City           str
State          str
Country        str
SellerID       str
OrderMonth     int32
OrderYear      int32
MonthName      str
DayName        str
Quarter        int32
PriceCategory  category
DiscountPercent float64
dtype: object
```

```
In [52]: df.head()
```

```
Out[52]:
```

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	C
48442	ORD0048443	2020-01-12	CUST010794	Anjali Patel	P00033	Puzzle 1000pc	I
81886	ORD0081887	2020-09-26	CUST031626	Vivaan Joshi	P00048	Wireless Charger	
4196	ORD0004197	2021-02-22	CUST022868	Kabir Gupta	P00029	T-Shirt	I
1929	ORD0001930	2023-08-18	CUST031209	Pooja Patel	P00025	Running Shoes	
58204	ORD0058205	2021-10-19	CUST033287	Priya Gupta	P00031	Kids Toy Car	I

5 rows × 27 columns



```
In [54]: df.to_csv("Amazon_Sales_cleaned1.csv", index=False)
```

```
In [55]: total_sales = df["TotalAmount"].sum()
print("Total_Sales:", total_sales)
```

Total_Sales: 91825647.91999999

```
In [56]: total_orders = df["OrderID"].count()
print("Total_Orders:", total_orders)
```

Total_Orders: 100000

```
In [57]: total_customers = df["CustomerID"].nunique()
print("Total_Customers:", total_customers)
```

Total_Customers: 43233

```
In [58]: average_order = df["TotalAmount"].mean()
print("Averag_Order:", average_order)
```

Averag_Order: 918.2564791999998

```
In [59]: max_order = df["TotalAmount"].max()
print("Highest Order Value:", max_order)
```

Highest Order Value: 3534.98

```
In [60]: min_order = df["TotalAmount"].min()
print("Lowest Order Value:", min_order)
```

Lowest Order Value: 4.27

```
In [61]: print("===== AMAZON SALES OVERVIEW =====")
print("Total Sales:", df["TotalAmount"].sum())
print("Total Orders:", df["OrderID"].count())
print("Unique Customers:", df["CustomerID"].nunique())
print("Average Order Value:", df["TotalAmount"].mean())
print("Highest Order Value:", df["TotalAmount"].max())
print("Lowest Order Value:", df["TotalAmount"].min())
```

===== AMAZON SALES OVERVIEW =====

Total Sales: 91825647.91999999

Total Orders: 100000

Unique Customers: 43233

Average Order Value: 918.2564791999998

Highest Order Value: 3534.98

Lowest Order Value: 4.27

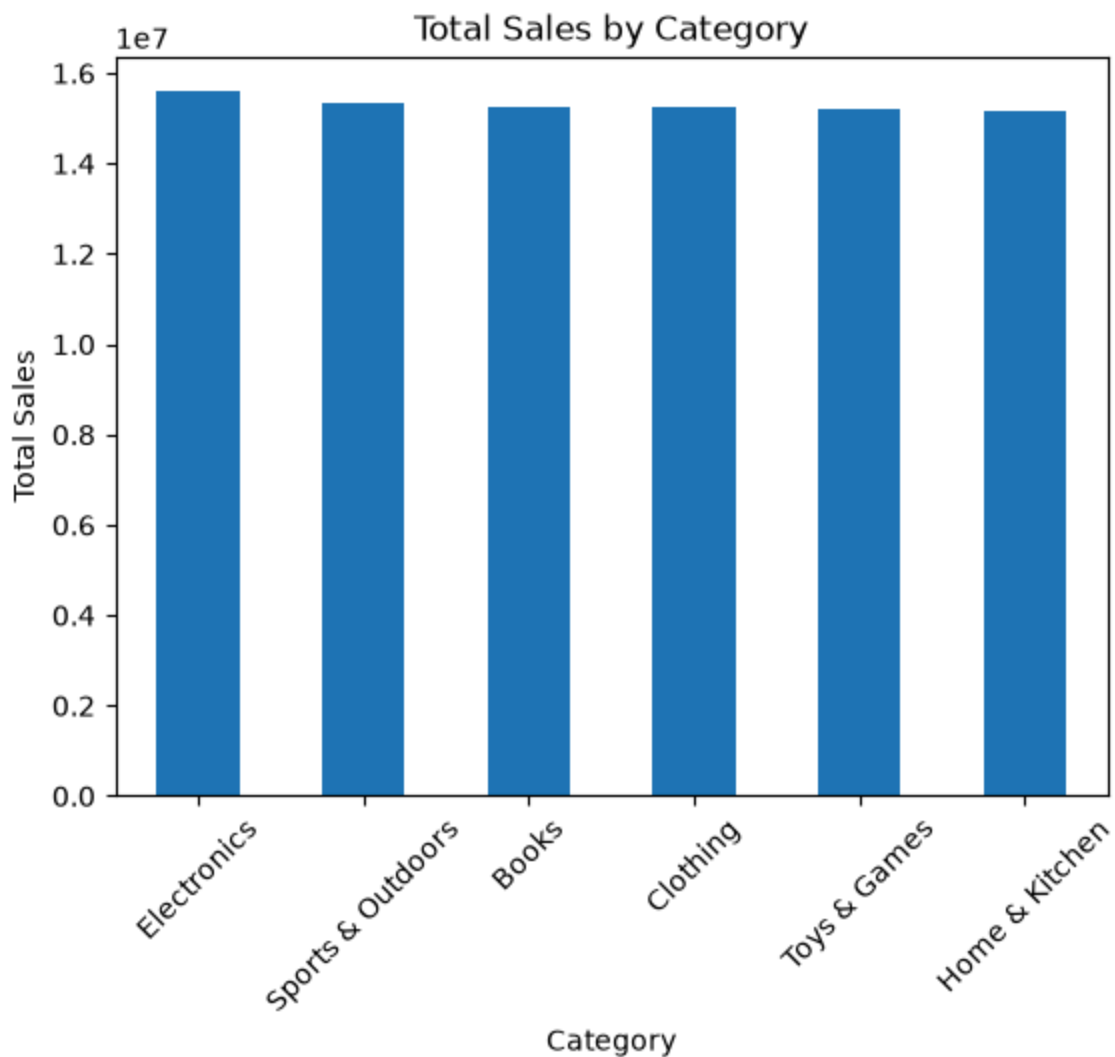
```
In [62]: category_sales = df.groupby("Category")["TotalAmount"].sum()
category_sales
```

```
Out[62]: Category
Books                15261837.01
Clothing             15253397.50
Electronics          15584217.18
Home & Kitchen        15163939.36
Sports & Outdoors     15345571.88
Toys & Games          15216684.99
Name: TotalAmount, dtype: float64
```

```
In [63]: category_sales = category_sales.sort_values(ascending=False)
category_sales
```

```
Out[63]: Category
Electronics      15584217.18
Sports & Outdoors 15345571.88
Books            15261837.01
Clothing         15253397.50
Toys & Games     15216684.99
Home & Kitchen   15163939.36
Name: TotalAmount, dtype: float64
```

```
In [64]: category_sales.plot(kind="bar")
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.savefig("Total Sales by Category.png")
plt.show()
```



```
In [65]: product_sales = df.groupby("ProductName")["TotalAmount"].sum()
product_sales
```

```

Out[65]:
ProductName
4K Monitor      1866774.35
Action Camera   1829029.24
Air Fryer       1789927.34
Backpack        1749971.14
Bluetooth Speaker 1803356.76
Board Game      1863310.96
Car Charger     1779795.27
Children's Book 1738112.57
Cookware Set    1877038.73
Desk Organizer  1843103.26
Desk Plant      1759672.62
Dress Shirt     1896765.36
Drone Mini      1816203.39
Electric Kettle 1905751.79
External HDD 2TB 1786108.33
Fitness Band    1784149.50
Gaming Mouse    1895103.98
Graphic Tablet  1786772.26
HDMI Cable 2m   1847856.25
Instant Pot     1838639.12
Jeans           1882591.82
Kids Toy Car    1891082.43
LED Desk Lamp   1921948.41
Laptop Sleeve   1848297.80
Mechanical Keyboard 1906963.54
Memory Card 128GB 1935138.40
Microphone      1871740.68
Noise Cancelling Headphones 1881108.57
Novel Bestseller 1803054.57
Office Chair    1827426.35
Phone Tripod    1866256.31
Portable SSD 1TB 1819377.38
Power Bank 20000mAh 1794738.58
Projector Mini  1805867.58
Puzzle 1000pc   1797812.49
Router          1850756.20
Running Shoes   1828658.19
Smart Light Bulb 1868940.58
Smartphone Case 1875421.41
Smartwatch      1901275.59
Sunglasses     1788111.81
T-Shirt         1787855.05
USB-C Charger   1763075.26
Vacuum Cleaner  1855117.83
Water Bottle    1895467.95
Webcam Full HD  1799332.82
Winter Jacket   1806053.51
Wireless Charger 1812628.92
Wireless Earbuds 1834207.58
Yoga Mat        1847898.09
Name: TotalAmount, dtype: float64

```

```

In [66]: product_sales = product_sales.sort_values(ascending=False)
product_sales

```

```

Out[66]: ProductName
Memory Card 128GB          1935138.40
LED Desk Lamp              1921948.41
Mechanical Keyboard        1906963.54
Electric Kettle            1905751.79
Smartwatch                 1901275.59
Dress Shirt                1896765.36
Water Bottle               1895467.95
Gaming Mouse               1895103.98
Kids Toy Car               1891082.43
Jeans                      1882591.82
Noise Cancelling Headphones 1881108.57
Cookware Set               1877038.73
Smartphone Case            1875421.41
Microphone                 1871740.68
Smart Light Bulb           1868940.58
4K Monitor                 1866774.35
Phone Tripod               1866256.31
Board Game                 1863310.96
Vacuum Cleaner             1855117.83
Router                     1850756.20
Laptop Sleeve              1848297.80
Yoga Mat                   1847898.09
HDMI Cable 2m              1847856.25
Desk Organizer             1843103.26
Instant Pot                1838639.12
Wireless Earbuds           1834207.58
Action Camera              1829029.24
Running Shoes              1828658.19
Office Chair               1827426.35
Portable SSD 1TB           1819377.38
Drone Mini                 1816203.39
Wireless Charger           1812628.92
Winter Jacket              1806053.51
Projector Mini             1805867.58
Bluetooth Speaker          1803356.76
Novel Bestseller           1803054.57
Webcam Full HD             1799332.82
Puzzle 1000pc              1797812.49
Power Bank 20000mAh        1794738.58
Air Fryer                  1789927.34
Sunglasses                 1788111.81
T-Shirt                    1787855.05
Graphic Tablet             1786772.26
External HDD 2TB           1786108.33
Fitness Band               1784149.50
Car Charger                1779795.27
USB-C Charger              1763075.26
Desk Plant                 1759672.62
Backpack                   1749971.14
Children's Book            1738112.57
Name: TotalAmount, dtype: float64

```

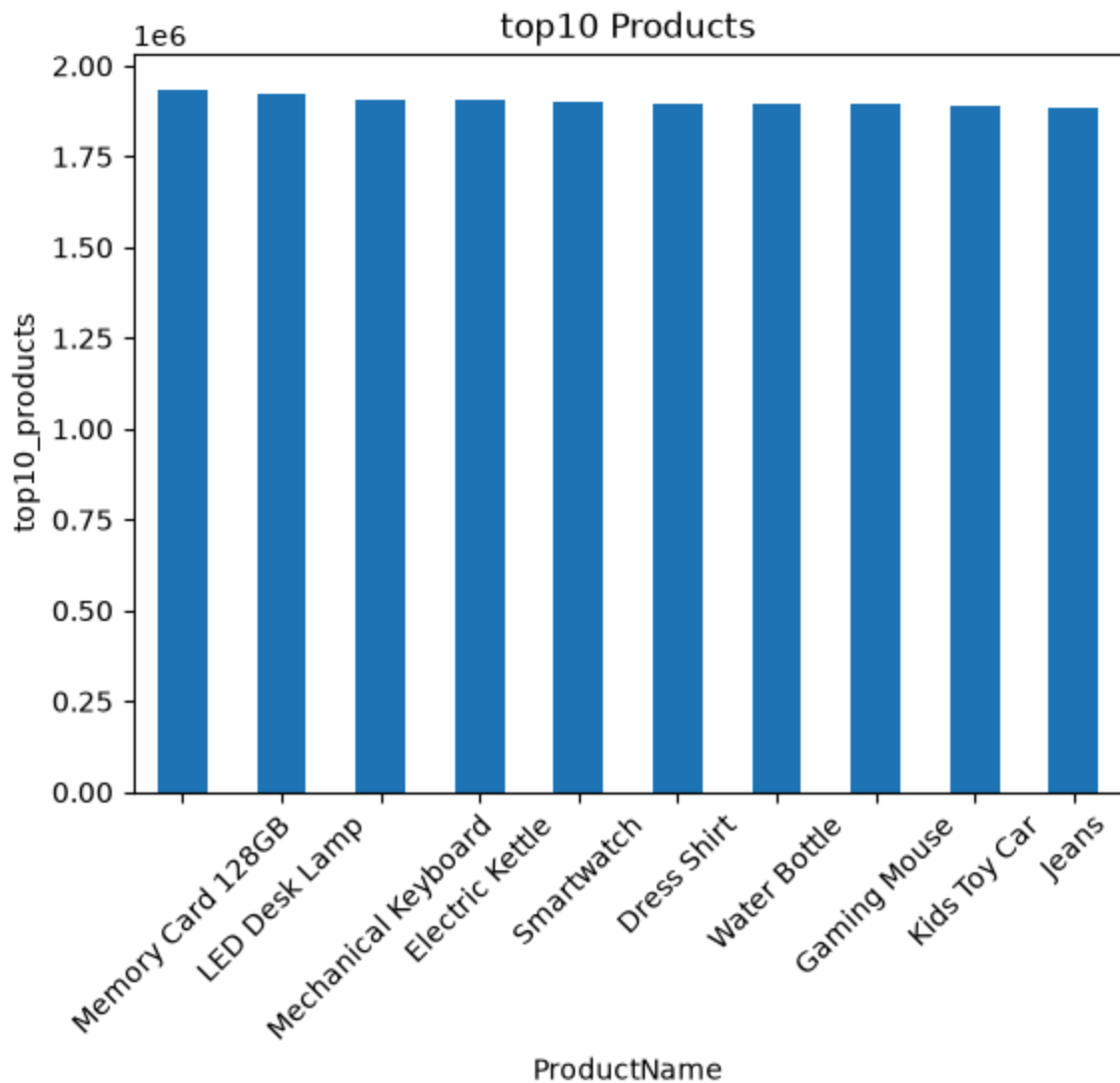
```

In [67]: top10_products = product_sales.head(10)
top10_products

```

```
Out[67]: ProductName
Memory Card 128GB      1935138.40
LED Desk Lamp          1921948.41
Mechanical Keyboard    1906963.54
Electric Kettle        1905751.79
Smartwatch             1901275.59
Dress Shirt            1896765.36
Water Bottle           1895467.95
Gaming Mouse           1895103.98
Kids Toy Car           1891082.43
Jeans                  1882591.82
Name: TotalAmount, dtype: float64
```

```
In [68]: top10_products.plot(kind = "bar")
plt.title("top10 Products")
plt.xlabel("ProductName")
plt.ylabel("top10_products")
plt.xticks(rotation=45)
plt.savefig("Top12 Products.png")
plt.show()
```



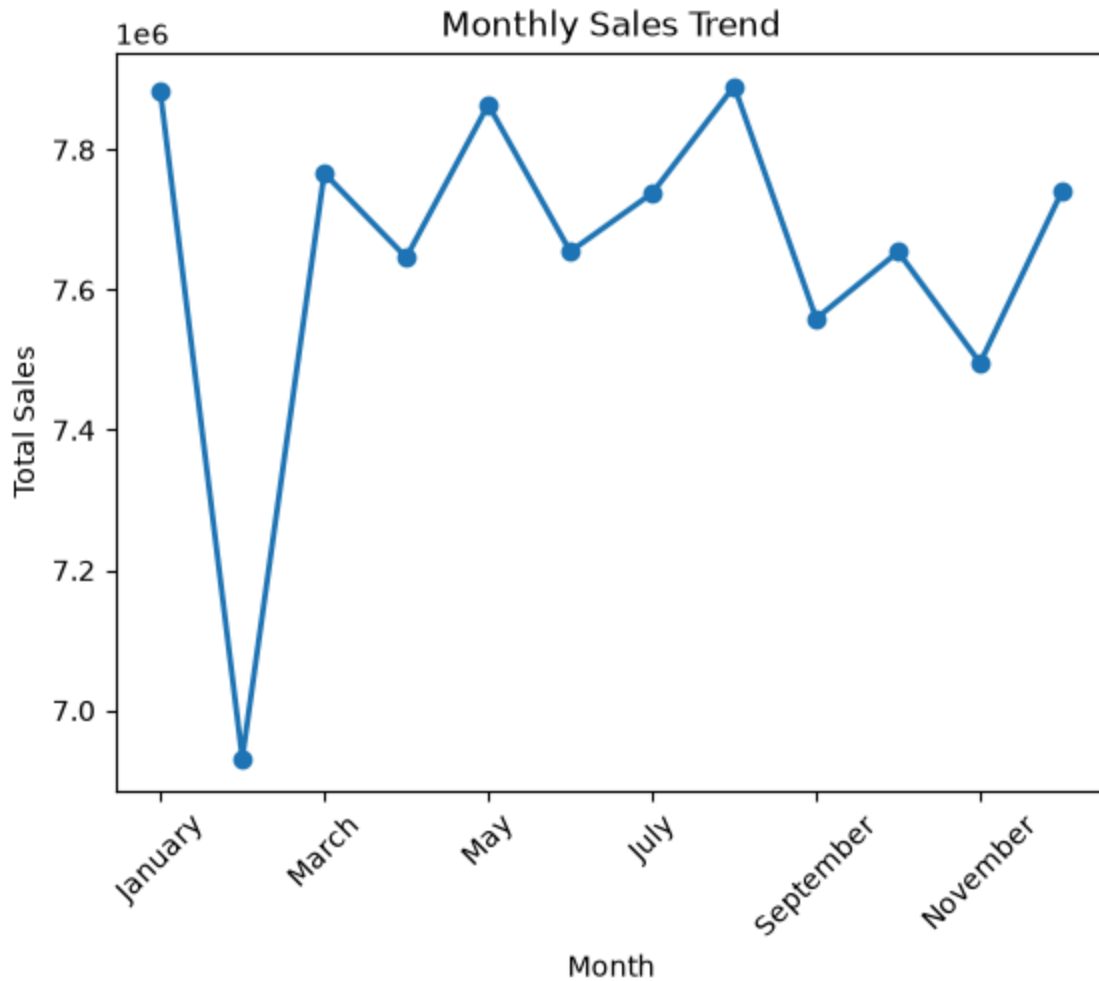
```
In [69]: monthly_sales = df.groupby("MonthName")["TotalAmount"].sum()  
monthly_sales
```

```
Out[69]: MonthName  
April      7646203.22  
August     7890598.37  
December   7740979.01  
February   6931724.49  
January    7883801.27  
July       7737925.51  
June       7654636.75  
March      7766527.73  
May        7864283.27  
November   7495754.13  
October    7654243.32  
September  7558970.85  
Name: TotalAmount, dtype: float64
```

```
In [70]: month_order=["January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"]  
monthly_sales = monthly_sales.reindex(month_order)  
monthly_sales
```

```
Out[70]: MonthName  
January      7883801.27  
February     6931724.49  
March        7766527.73  
April        7646203.22  
May          7864283.27  
June         7654636.75  
July         7737925.51  
August       7890598.37  
September    7558970.85  
October      7654243.32  
November     7495754.13  
December     7740979.01  
Name: TotalAmount, dtype: float64
```

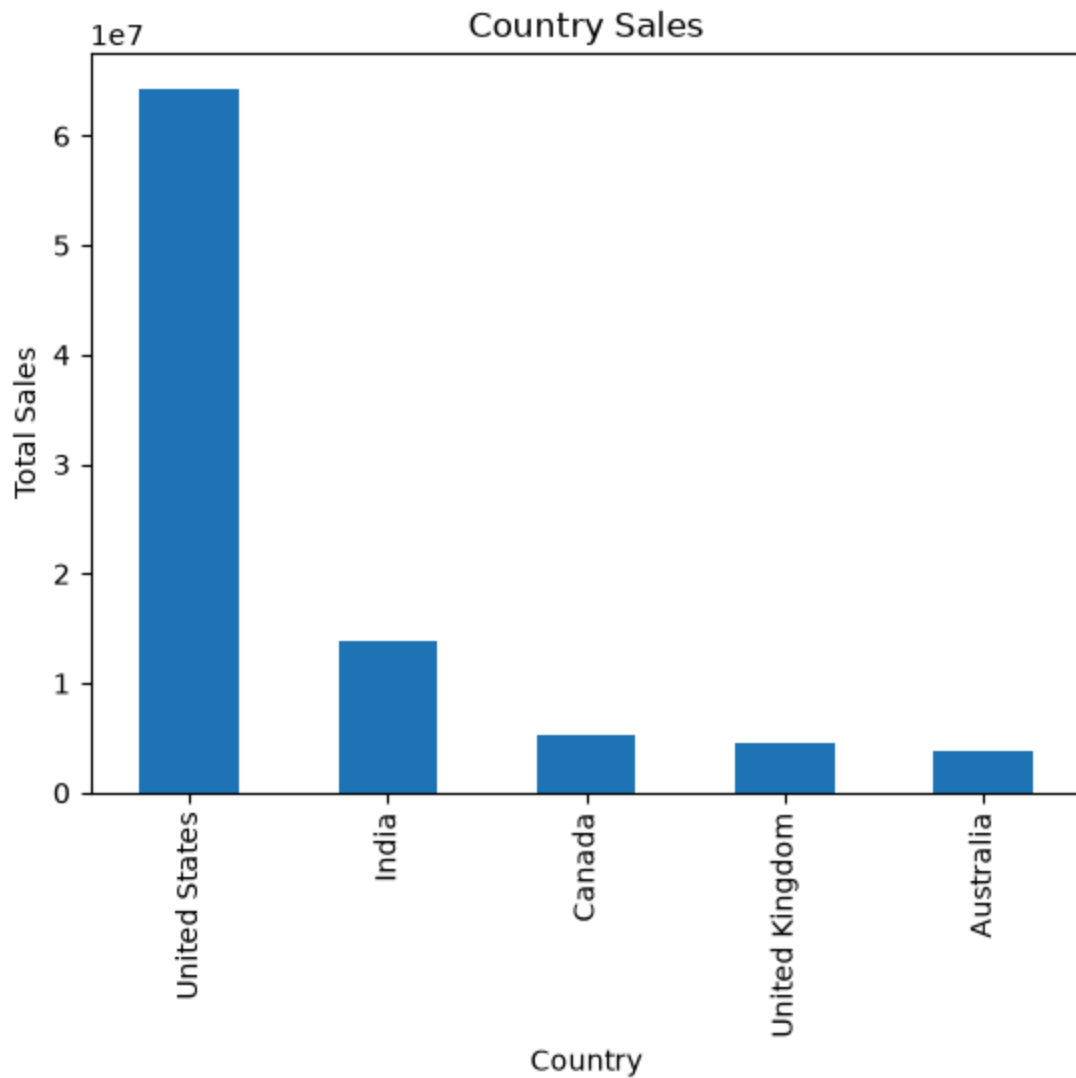
```
In [71]: monthly_sales.plot(kind="line", marker="o", linewidth=2)  
plt.title("Monthly Sales Trend")  
plt.xlabel("Month")  
plt.ylabel("Total Sales")  
plt.xticks(rotation=45)  
plt.savefig("Monthly Sales Trend.png")  
plt.show()
```



```
In [72]: country_sales = df.groupby("Country")["TotalAmount"].sum()
country_sales = country_sales.sort_values(ascending=False)
country_sales
```

```
Out[72]: Country
United States    64310048.50
India            13875839.12
Canada           5323757.00
United Kingdom   4526896.86
Australia        3789106.44
Name: TotalAmount, dtype: float64
```

```
In [73]: country_sales.plot(kind = "bar")
plt.title("Country Sales")
plt.xlabel("Country")
plt.ylabel("Total Sales")
plt.savefig("Country Sales.png")
plt.show()
```



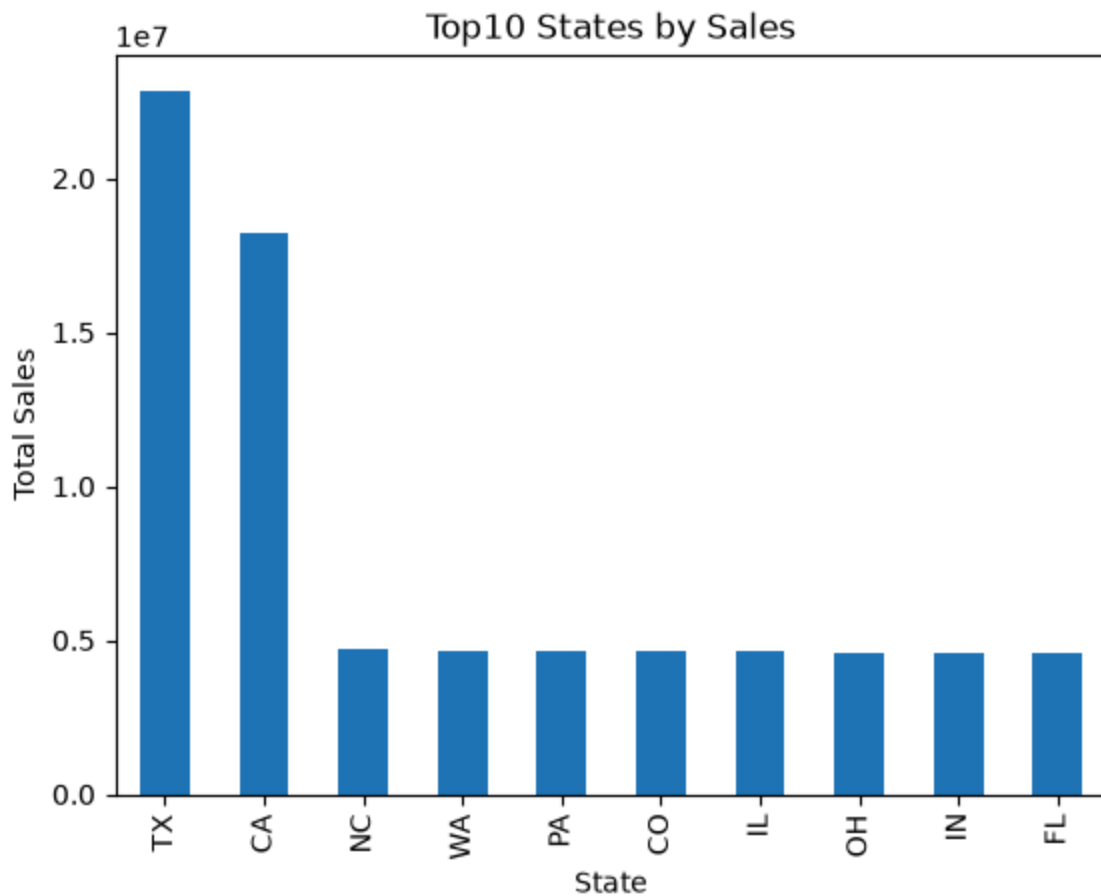
```
In [74]: state_sales = df.groupby("State")["TotalAmount"].sum()
state_sales = state_sales.sort_values(ascending=False)
state_sales
```

```
Out[74]: State
TX      22862540.54
CA      18231196.79
NC       4747701.73
WA       4660962.17
PA       4650673.88
CO       4638498.97
IL       4632847.44
OH       4615442.32
IN       4609048.43
FL       4597177.27
NY       4554358.08
DC       4520876.11
AZ       4504324.19
Name: TotalAmount, dtype: float64
```

```
In [75]: top10_states = state_sales.head(10)
top10_states
```

```
Out[75]: State
TX      22862540.54
CA      18231196.79
NC       4747701.73
WA      4660962.17
PA      4650673.88
CO      4638498.97
IL      4632847.44
OH      4615442.32
IN      4609048.43
FL      4597177.27
Name: TotalAmount, dtype: float64
```

```
In [76]: top10_states.plot(kind = "bar")
plt.title("Top10 States by Sales")
plt.xlabel("State")
plt.ylabel("Total Sales")
plt.savefig("Top10 State Sales.png")
plt.show()
```



```
In [77]: print(country_sales.idxmax())
```

United States

```
In [78]: print(state_sales.idxmax())
```

TX

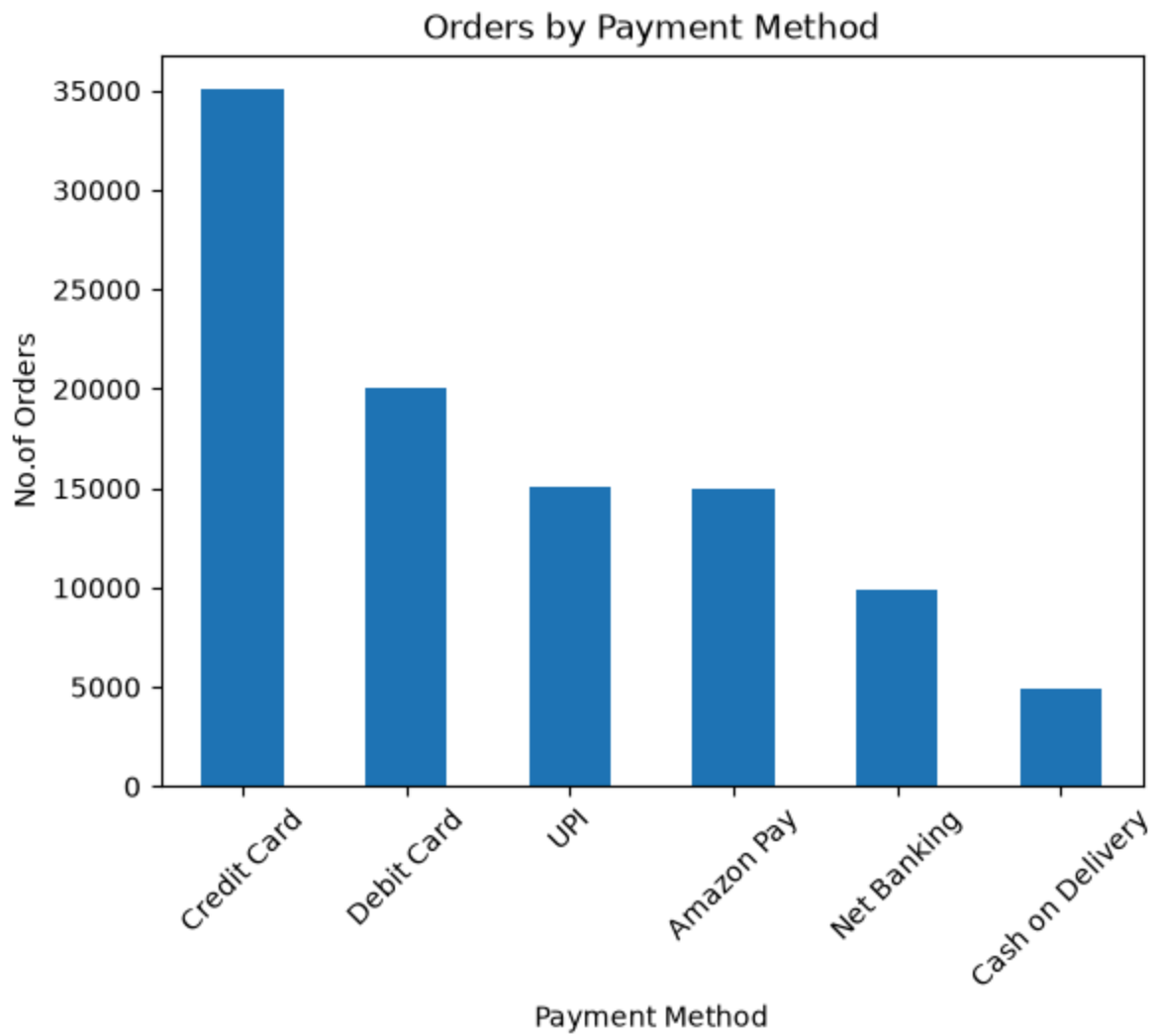
```
In [79]: print(country_sales.max())  
print(state_sales.max())
```

```
64310048.5  
22862540.54
```

```
In [80]: payment_count = df["PaymentMethod"].value_counts()  
payment_count
```

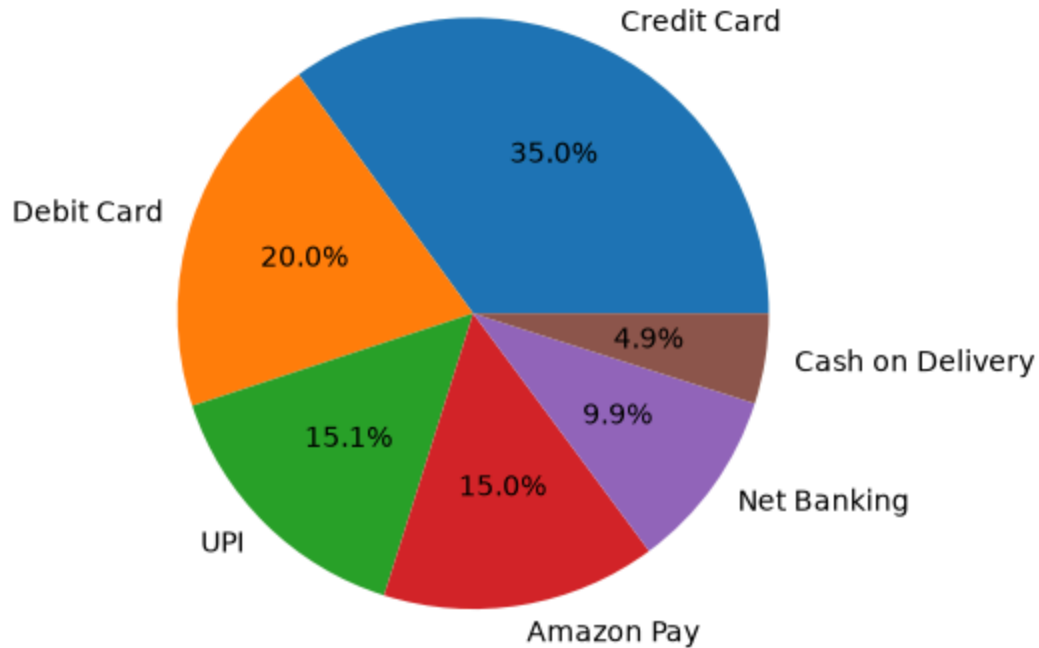
```
Out[80]: PaymentMethod  
Credit Card      35038  
Debit Card       20024  
UPI              15066  
Amazon Pay       15017  
Net Banking      9927  
Cash on Delivery 4928  
Name: count, dtype: int64
```

```
In [81]: payment_count.plot(kind="bar")  
plt.title("Orders by Payment Method")  
plt.xlabel("Payment Method")  
plt.ylabel("No.of Orders")  
plt.xticks(rotation=45)  
plt.savefig("Payment Method.png")  
plt.show()
```



```
In [82]: payment_count.plot(kind = "pie", autopct = "%1.1f%")  
plt.ylabel("")  
plt.title("Payment Method Distribution")  
plt.savefig("Payment Method Distribution.png")  
plt.show()
```

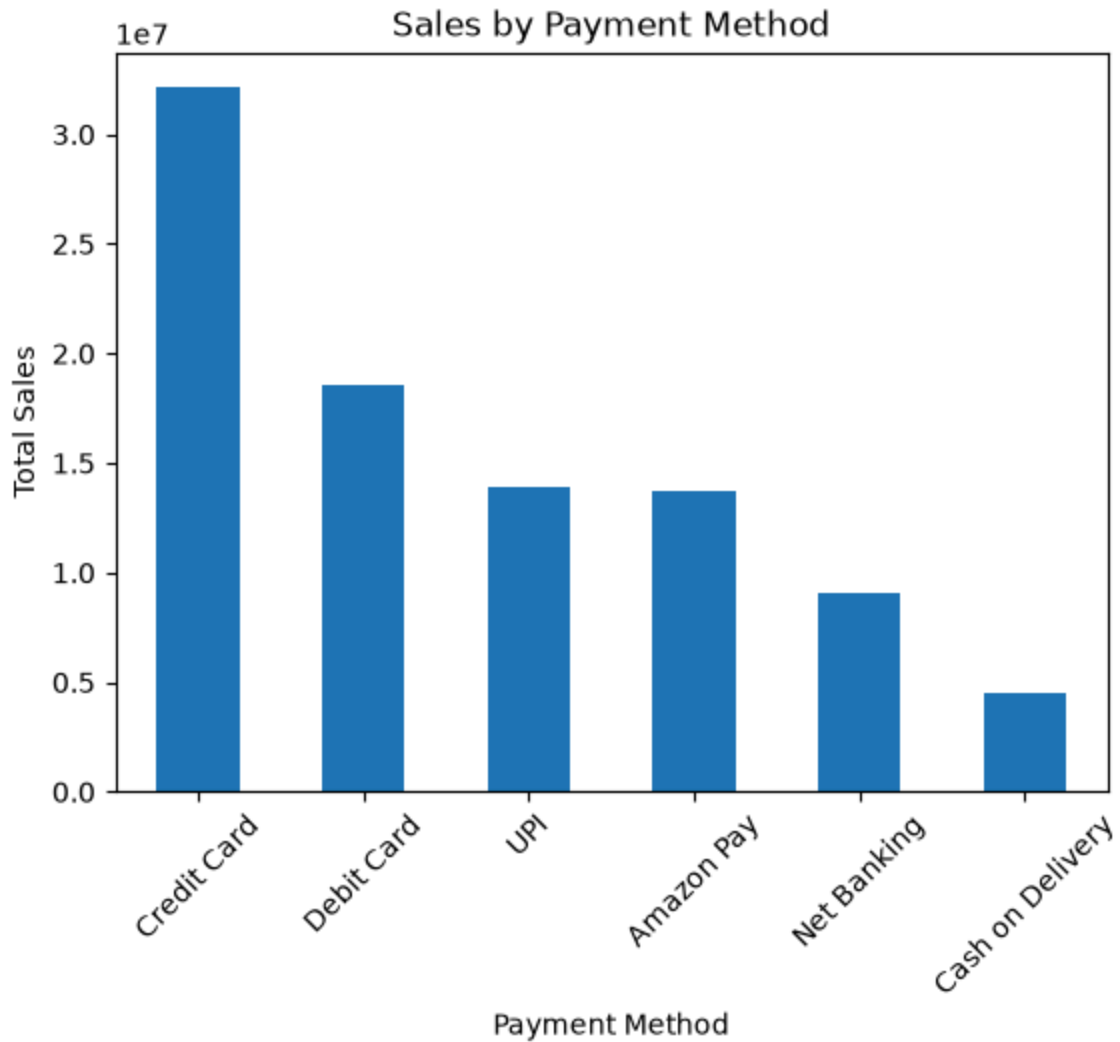
Payment Method Distribution



```
In [83]: payment_sales = df.groupby("PaymentMethod")["TotalAmount"].sum()  
payment_sales = payment_sales.sort_values(ascending=False)  
payment_sales
```

```
Out[83]: PaymentMethod  
Credit Card      32122158.69  
Debit Card       18538678.53  
UPI              13896028.55  
Amazon Pay       13697498.42  
Net Banking      9055674.57  
Cash on Delivery  4515609.16  
Name: TotalAmount, dtype: float64
```

```
In [84]: payment_sales.plot(kind="bar")  
plt.title("Sales by Payment Method")  
plt.xlabel("Payment Method")  
plt.ylabel("Total Sales")  
plt.xticks(rotation=45)  
plt.savefig("Payment Method Sales.png")  
plt.show()
```



```
In [85]: print(payment_count.idxmax())
```

Credit Card

```
In [86]: print(payment_sales.idxmax())
```

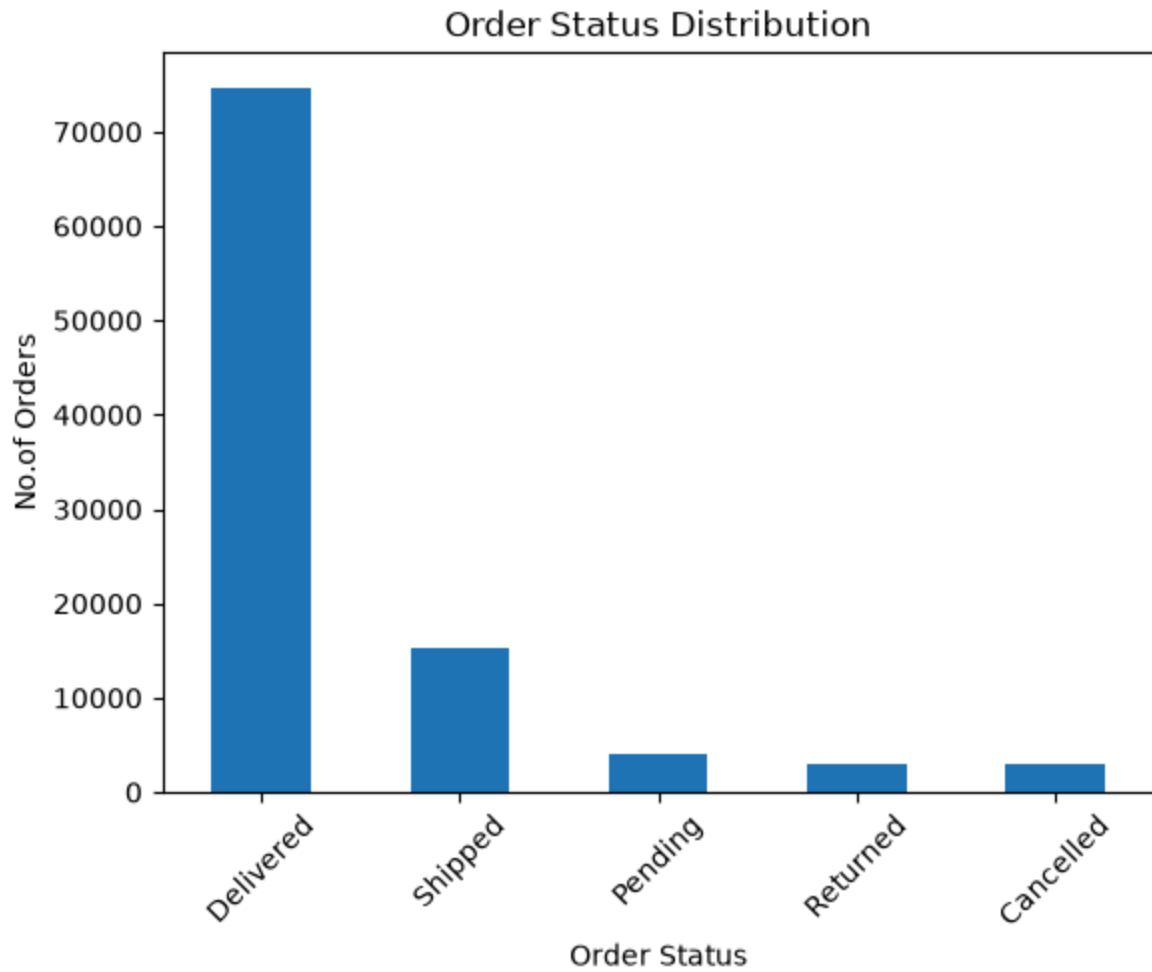
Credit Card

```
In [87]: order_status = df["OrderStatus"].value_counts()
order_status
```

```
Out[87]: OrderStatus
Delivered    74628
Shipped      15192
Pending       4103
Returned      3049
Cancelled     3028
Name: count, dtype: int64
```

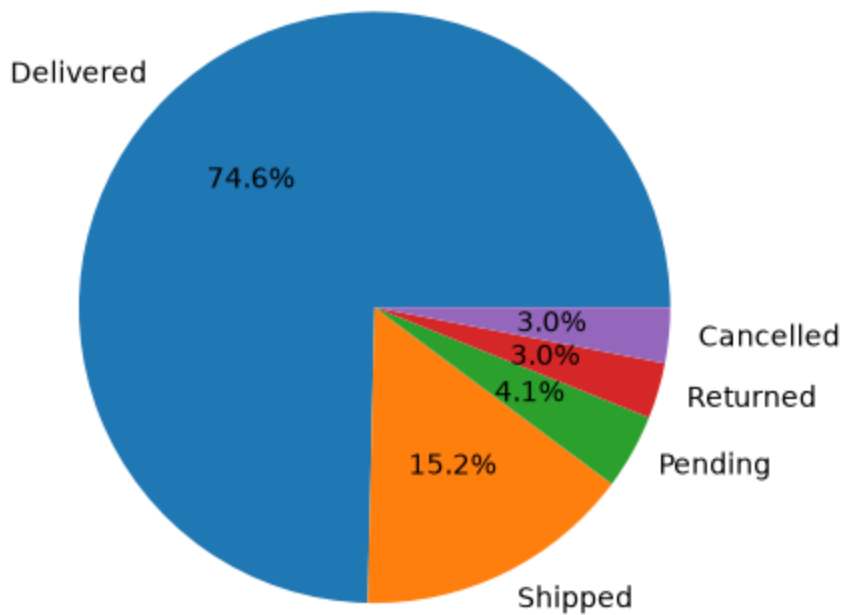
```
In [88]: order_status.plot(kind="bar")
plt.title("Order Status Distribution")
plt.xlabel("Order Status")
plt.ylabel("No.of Orders")
plt.xticks(rotation=45)
```

```
plt.savefig("Payment Method.png")
plt.show()
```



```
In [89]: order_status.plot(kind = "pie", autopct = "%1.1f%")
plt.ylabel("")
plt.title("Order Status Distribution")
plt.savefig("Order Status Pie_Distribution.png")
plt.show()
```

Order Status Distribution



```
In [90]: print(order_status.idxmax())
```

Delivered

```
In [91]: print(order_status.idxmin())
```

Cancelled

```
In [92]: print(order_status.max())
```

74628

```
In [93]: print(order_status.min())
```

3028

```
In [96]: average_discount = df["DiscountPercent"].mean()
print("Aerage Discount:", average_discount)
```

Aerage Discount: 7.4226

```
In [97]: max_discount = df["DiscountPercent"].max()
print("Maximum Discount:", max_discount)
```

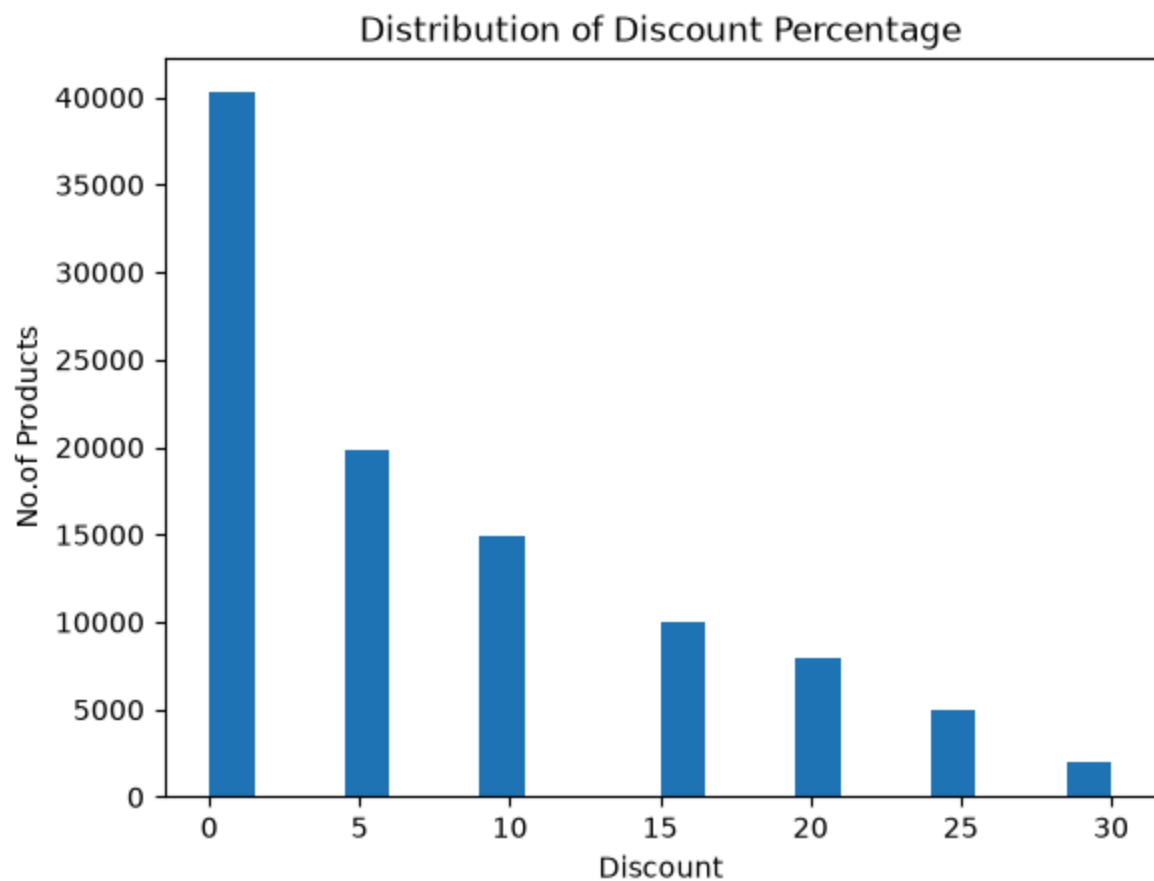
Maximum Discount: 30.0

```
In [98]: min_discount = df["DiscountPercent"].min()
print("Minimum Discount:", min_discount)
```

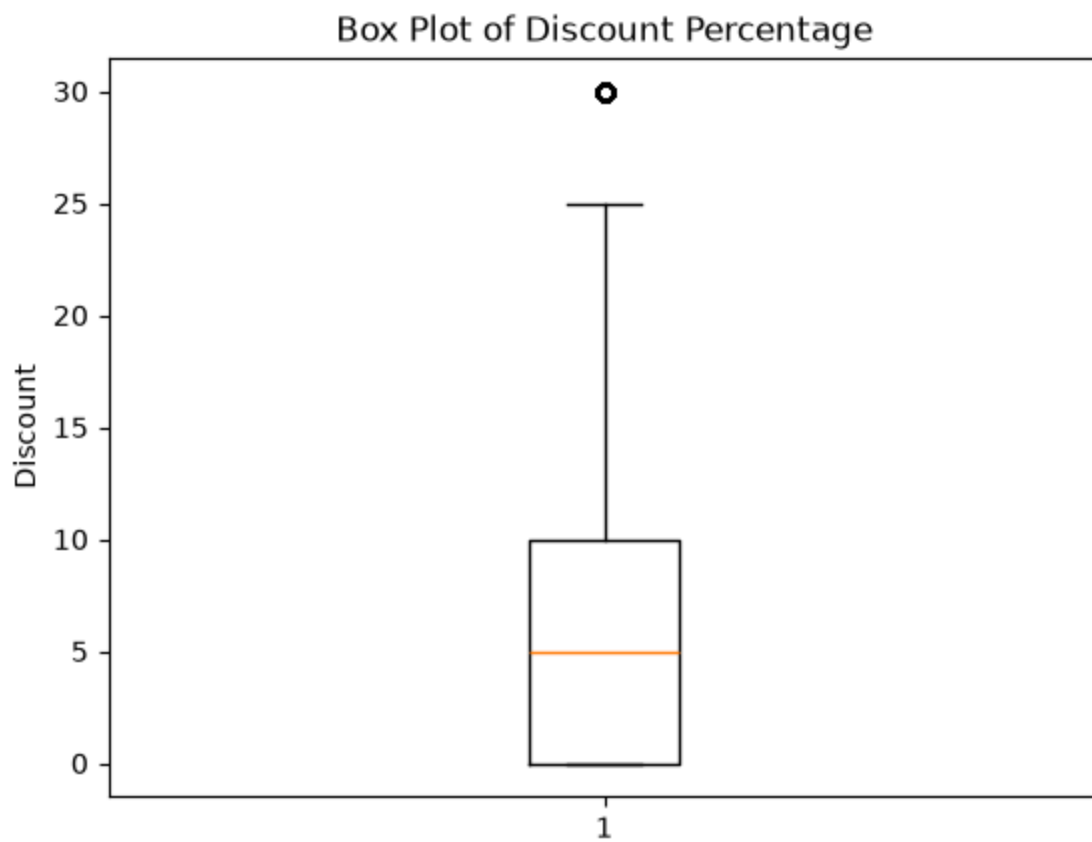
Minimum Discount: 0.0

```
In [100...]: plt.hist(df["DiscountPercent"], bins=20)
plt.title("Distribution of Discount Percentage")
plt.xlabel("Discount")
```

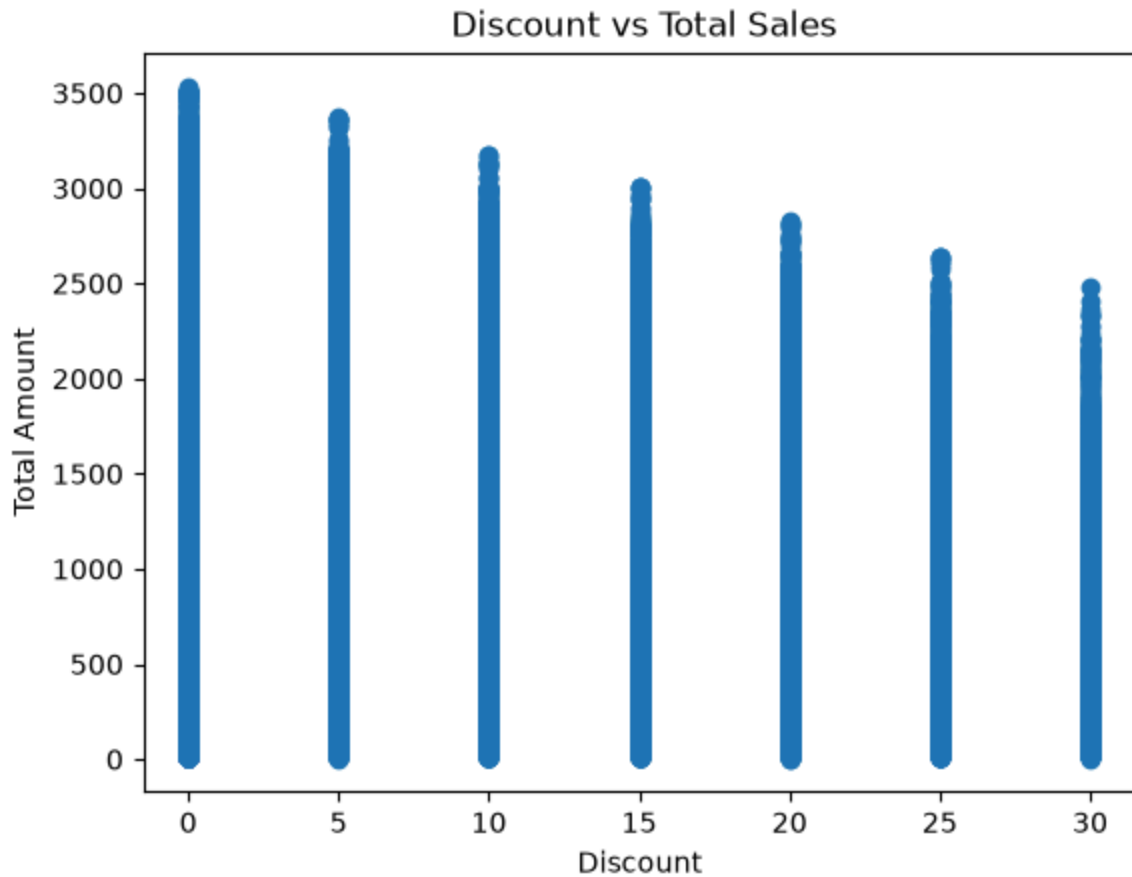
```
plt.ylabel("No.of Products")  
plt.show()
```



```
In [101... plt.boxplot(df["DiscountPercent"])  
plt.title("Box Plot of Discount Percentage")  
plt.ylabel("Discount")  
plt.show()
```



```
In [102... plt.scatter(df["DiscountPercent"], df["TotalAmount"])
plt.title(" Discount vs Total Sales")
plt.xlabel("Discount")
plt.ylabel("Total Amount")
plt.show()
```



```
In [103... average_shipping = df["ShippingCost"].mean()
print("Average Shipping Cost:", average_shipping)
```

Average Shipping Cost: 7.4066601

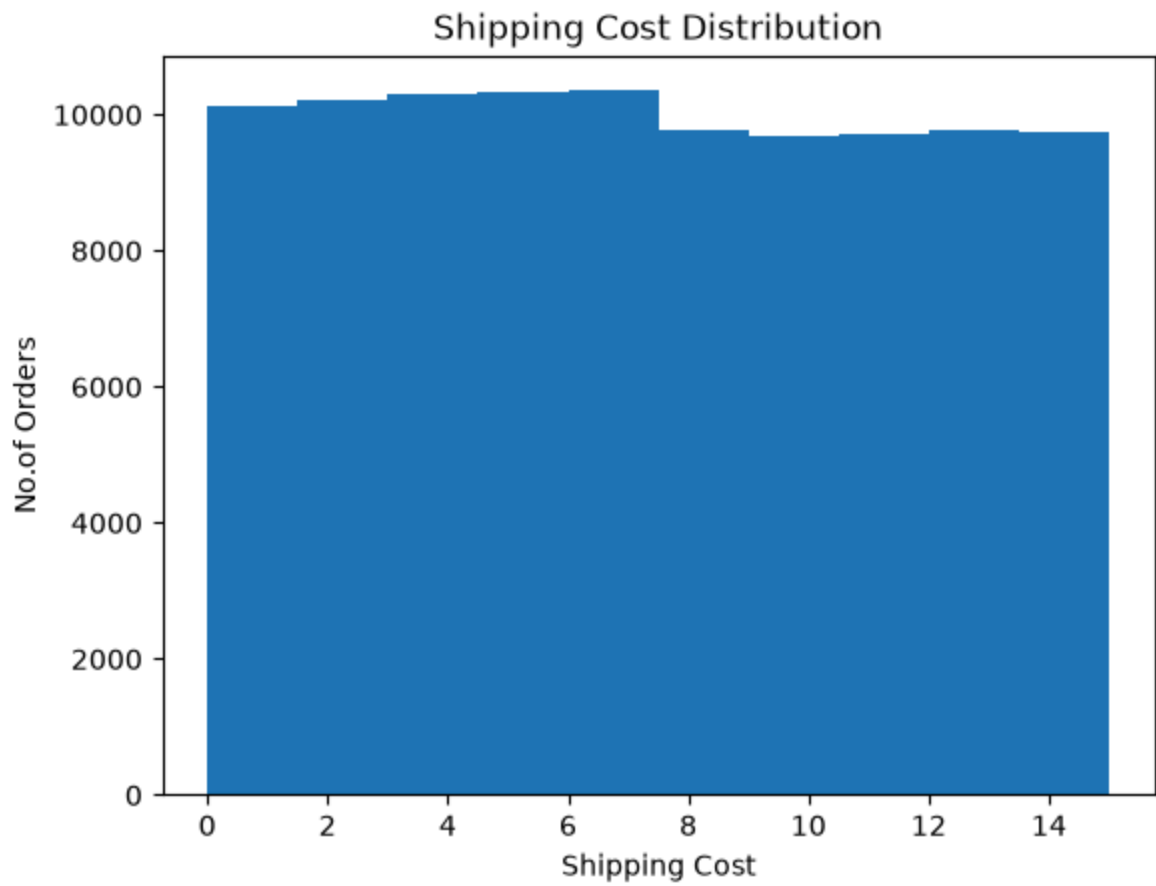
```
In [104... max_shipping = df["ShippingCost"].max()
print("Maximum Shipping Cost:", max_shipping)
```

Maximum Shipping Cost: 15.0

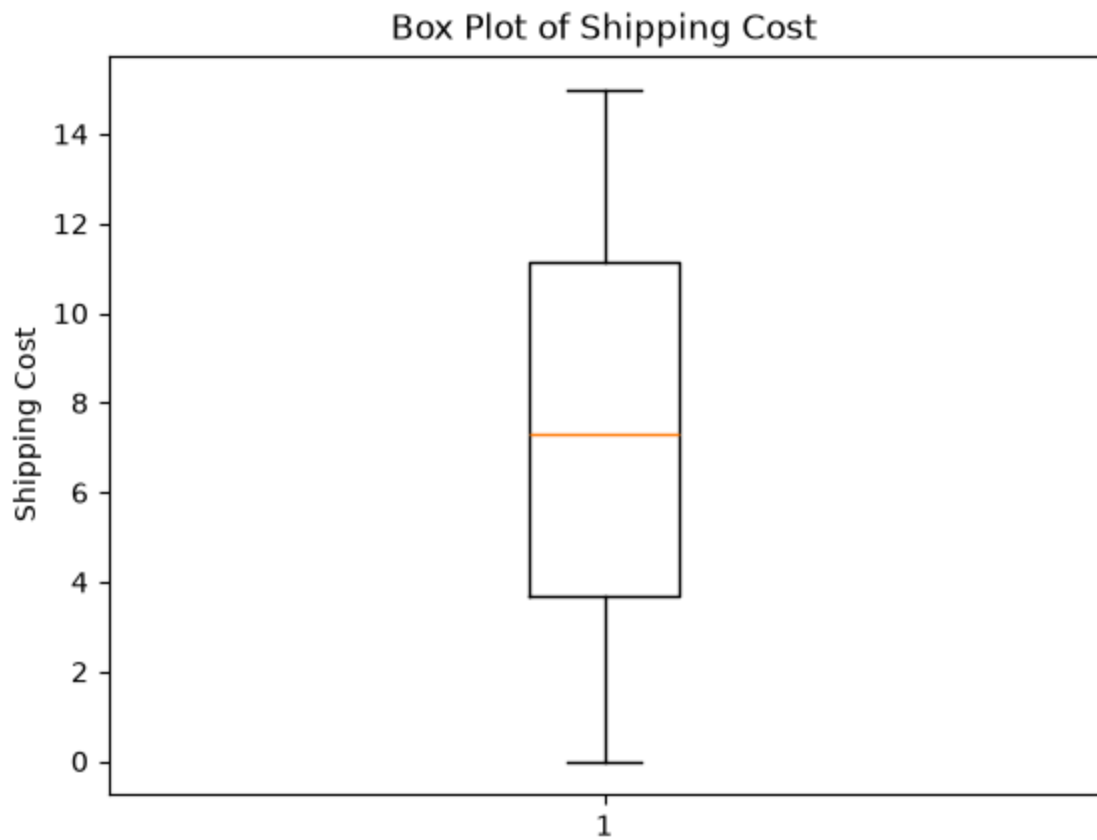
```
In [105... min_shipping = df["ShippingCost"].min()
print("Minimum Shipping Cost:", min_shipping)
```

Minimum Shipping Cost: 0.0

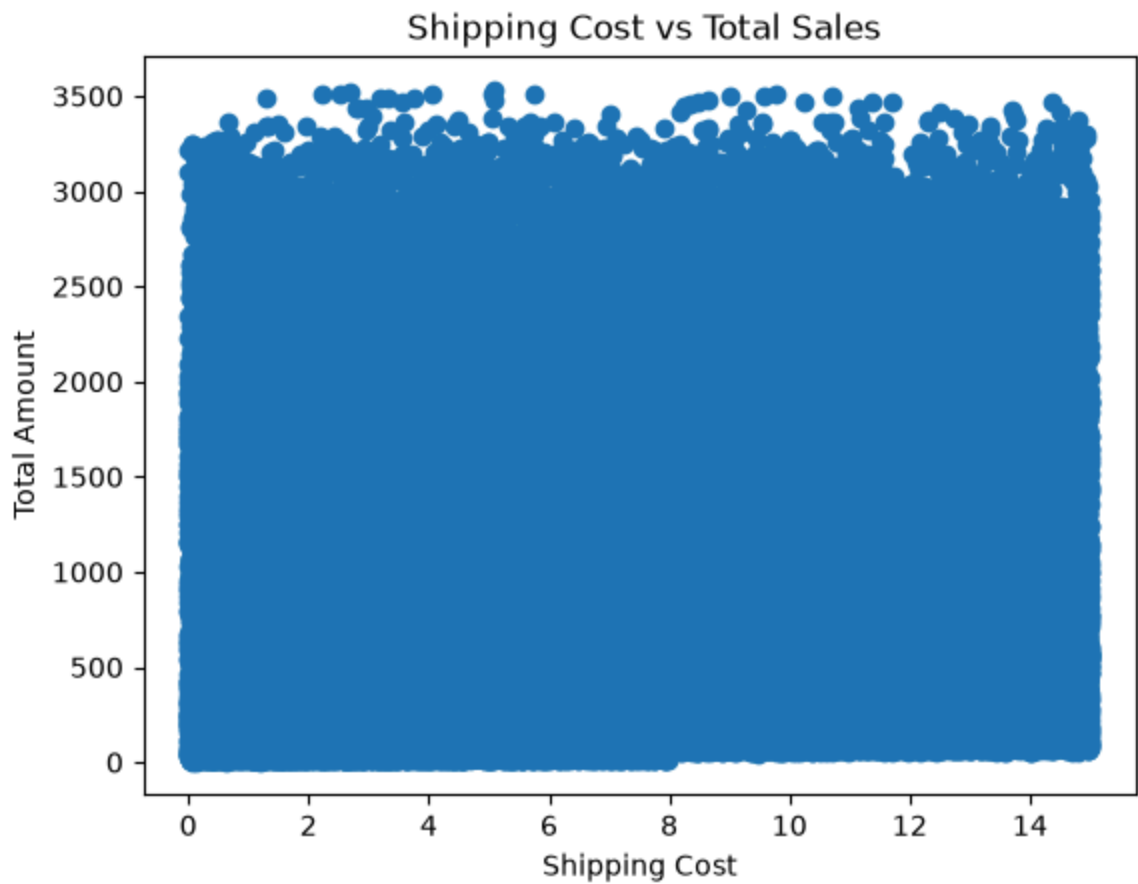
```
In [107... plt.hist(df["ShippingCost"], bins=10)
plt.title("Shipping Cost Distribution")
plt.xlabel("Shipping Cost")
plt.ylabel("No.of Orders")
plt.show()
```



```
In [108... plt.boxplot(df["ShippingCost"])  
plt.title("Box Plot of Shipping Cost")  
plt.ylabel("Shipping Cost")  
plt.show()
```



```
In [109... plt.scatter(df["ShippingCost"], df["TotalAmount"])
plt.title(" Shipping Cost vs Total Sales")
plt.xlabel("Shipping Cost")
plt.ylabel("Total Amount")
plt.show()
```



In [111...

```
numerical_df = df.select_dtypes(include=["int64","float64"])
numerical_df
```

Out[111...

	Quantity	UnitPrice	Discount	Tax	ShippingCost	TotalAmount	DiscountPercent
48442	5	598.29	0.00	538.46	5.07	3534.98	0.0
81886	5	596.47	0.00	536.82	2.68	3521.85	0.0
4196	5	595.26	0.00	535.73	2.21	3514.24	0.0
1929	5	594.85	0.00	535.36	4.04	3513.65	0.0
58204	5	594.46	0.00	535.01	5.74	3513.05	0.0
...	...	...	...	...	...	...	...
38172	1	5.89	0.00	0.47	0.05	6.41	0.0
4730	1	6.25	0.20	0.00	1.19	6.19	20.0
99805	1	6.08	0.05	0.29	0.12	6.19	5.0
68176	1	5.26	0.20	0.76	0.62	5.59	20.0
86771	1	5.57	0.30	0.31	0.06	4.27	30.0

100000 rows × 7 columns



```
In [113... correlation = numerical_df.corr()  
correlation
```

Out[113...

	Quantity	UnitPrice	Discount	Tax	ShippingCost	TotalAmount	DiscountPercent
Quantity	1.000000	0.004408	-0.000398	0.437884	0.020047	0.594928	
UnitPrice	0.004408	1.000000	0.006159	0.526856	0.045773	0.715808	
Discount	-0.000398	0.006159	1.000000	-0.077550	0.000315	-0.108088	
Tax	0.437884	0.526856	-0.077550	1.000000	0.023498	0.779399	
ShippingCost	0.020047	0.045773	0.000315	0.023498	1.000000	0.039349	
TotalAmount	0.594928	0.715808	-0.108088	0.779399	0.039349	1.000000	
DiscountPercent	-0.000398	0.006159	1.000000	-0.077550	0.000315	-0.108088	

```
In [117... sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")  
plt.title("Correlation Heatmap")  
plt.savefig("corr_heatmap.png")  
plt.show()
```

